Table 1.9 Aramid fibres

Fibre	Properties	Uses
Kevlar®	Lightweight, flexible and extremely durable aramid; excellent resistance to heat, corrosion and damage from chemicals and high tensile strength-to-weight ratio	Used in protective clothing such as police body armour, where the fibre is woven in a lattice that provides protection against knife attack
Nomex®	Extremely strong synthetic fabric that can withstand exposure to the most extreme conditions	Primarily used where resistance to heat and flames is essential, for example firefighters' uniforms

Sports textiles

- **Micro-encapsulation** is a process of applying microscopic capsules to fibres and fabrics.
- The capsules can contain vitamins, therapeutic oils, moisturisers, antiseptics and antibacterial chemicals that are released through friction.
- Rhovyl® is a non-flammable, synthetic fibre that is crease resistant, has good thermal and acoustic properties, is antibacterial and comfortable to wear.
- The construction of the fibre gives fabrics the ability to wick away moisture, such as perspiration, through the fabric. It also dries quickly, meaning it does not retain odours, making it ideal for socks and sportswear.

Micro-encapsulation: tiny microscopic droplets containing various substances applied to fibres, yarns and materials including paper and card.

Now test yourself

TESTED

1 Describe what gives plywood its strength. (2)
2 Explain why carbon fibre reinforced polymer (CFRP) is a suitable material for a racing bike. (3)
3 Explain how micro-encapsulation is beneficial to a patient when used in medical dressings. (2)
4 State what is meant by 'smart glass'. (1)
5 Describe **two** reasons why the aramid fibre Nomex® is used in protective clothing. (2)

1.5 Mechanical devices

1.5.1 Types of movement

REVISED

The four main types of movement are:

- **linear**: movement in a straight line
- **reciprocation**: movement that goes back and forth in a straight line
- **rotary**: movement that turns in a circle
- **oscillation**: a swinging movement from side to side.

1.5.2 Classification of levers

REVISED

- **Levers** consist of a rod pivoted on a **fulcrum**.
- In a class 1 lever, the effort and the load are on opposite sides of the fulcrum. An example is a see-saw.
- In a class 2 lever, the load and the effort are on the same side of the fulcrum, with the load nearest the fulcrum. The effort required is less than the load. An example is a wheelbarrow.
- In a class 3 lever, the load and effort are on the same side of the fulcrum, but this time the effort is nearest to it. The effort required is greater than the load, but the load can be moved faster. An example is a pair of tweezers.

Lever: a rod pivoted on a fulcrum – the class of lever (1, 2 or 3) affects how much effort is needed to move the load.

Fulcrum: the point, or pivot, on which a lever sits.

Exam tip

Make sure that you show all of your working when performing calculations related to mechanical systems, including writing down the formula.

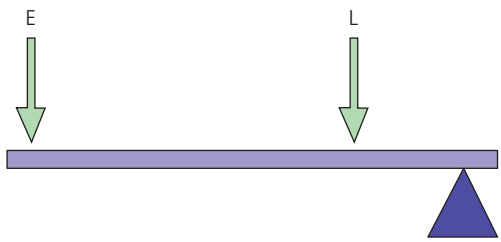


Figure 1.11 A class 2 lever

$$\text{Mechanical advantage of a lever} = \frac{\text{Load } (F_b)}{\text{Effort } (F_a)}$$

$$\text{Velocity ratio of a lever} = \frac{\text{Distance moved by effort}}{\text{Distance moved by load}}$$

$$\text{Efficiency of a lever} = \frac{\text{Work done on the load}}{\text{Work done by the effort}} \times 100$$

1.5.3 Linkages

REVISED

- **Linkages** are used to change the size of a force, the direction of motion and/or the type of motion.
- In a **reverse motion linkage**, the bottom link moves to the left as the top link moves to the right. If the distance between the fixed pivot and the moving pivots is equal, then the output force is the same as the input force. The output force can be increased or decreased by changing the position of the fixed pivot.
- In a **bell crank** the output movement is at 90 degrees to the input movement. The output force is greater than the input force when the fixed pivot is closer to the output lever.

Linkage: levers connected together via fixed and moving pivots; they are used to change the size of a force and/or the direction of motion.

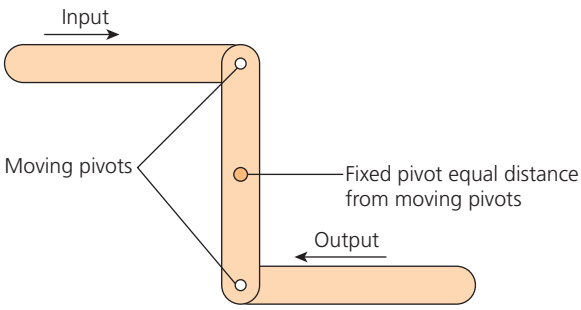


Figure 1.12 A reverse motion linkage

1.5.4 Cams

REVISED

- **Cams and followers** turn rotary motion into reciprocating motion. The follower moves up and down as the cam rotates.
- With a **pear-shaped cam**, the follower remains stationary (dwells) for half of the cycle. It is then pushed up as the point of the cam approaches. Finally, as the point passes, the follower falls and then dwells again.
- **Eccentric cams** are circular but have an off-centre rotating shaft. The rise and fall produced is symmetrical.
- With a **snail/drop cam** the follower dwells for the first 120 degrees or so of the cycle. It then rises slowly before dropping suddenly when it reaches the peak of the snail shape.

Cam and follower: a mechanism that converts rotary motion into reciprocating motion.

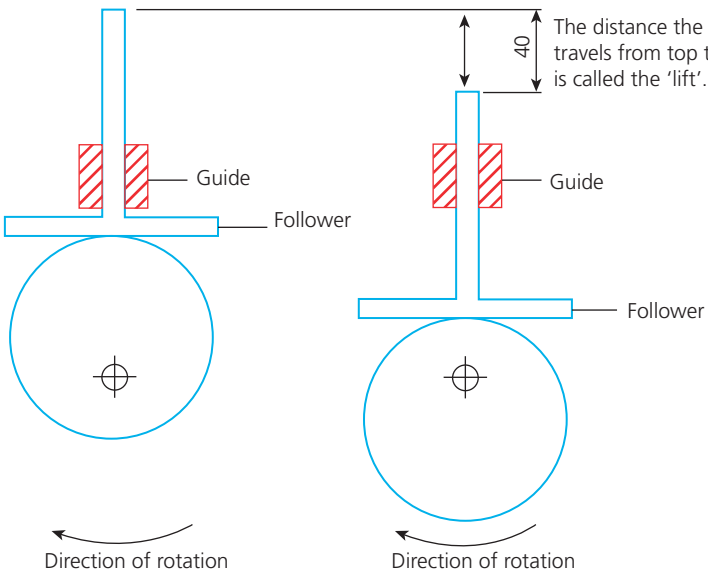


Figure 1.13 An eccentric cam mechanism

1.5.5 Followers

REVISED

- **Knife followers** have a sharp point that makes contact with the cam. They often wear quickly, leading to the mechanism not working correctly.
- **Roller followers** replace the sliding motion between the cam and follower with a rolling motion. This reduces wear.
- **Flat followers** are used where space is limited.

1.5.6 Pulleys and belts

REVISED

- V-belt pulley systems transmit rotary motion and torque. A larger pulley driving a smaller pulley results in increased speed but decreased torque. A smaller pulley driving a larger pulley achieves the opposite.
- The speed of a pulley is measured in revolutions per minute (rpm).

Pulley and belt: A system that uses pulley wheels and a belt to transmit rotary motion and torque.

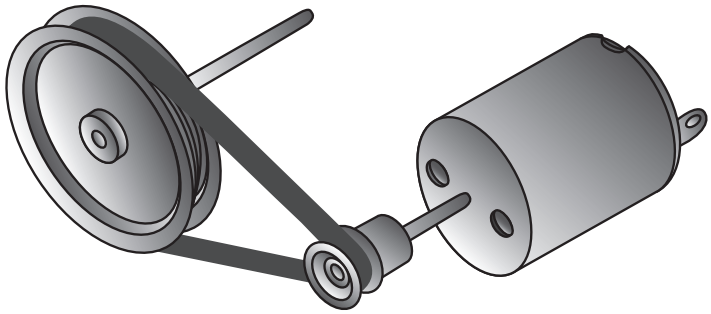


Figure 1.14 An electric motor driving a simple pulley system

Typical mistake

It is easy to mix up the driven pulley and the driver pulley when calculating velocity ratios. Ensure you know which one is which.

$$\text{Velocity ratio (VR) of a pulley system} = \frac{\text{Diameter of the driven pulley}}{\text{Diameter of the driver pulley}}$$

$$\text{Speed of the driven pulley (rpm)} = \frac{\text{rpm of the driver pulley}}{\text{Velocity ratio}}$$

1.5.7 Cranks and sliders

REVISED

- **Crank and slider** mechanisms convert rotary motion to reciprocating motion, and vice versa. To create reciprocating motion, the crank rotates and the connecting rod pushes the slider backwards and forwards.
- Piston engines in cars are an example of the reverse.

Crank and slider: a mechanism that converts rotary motion to reciprocating motion and vice versa.

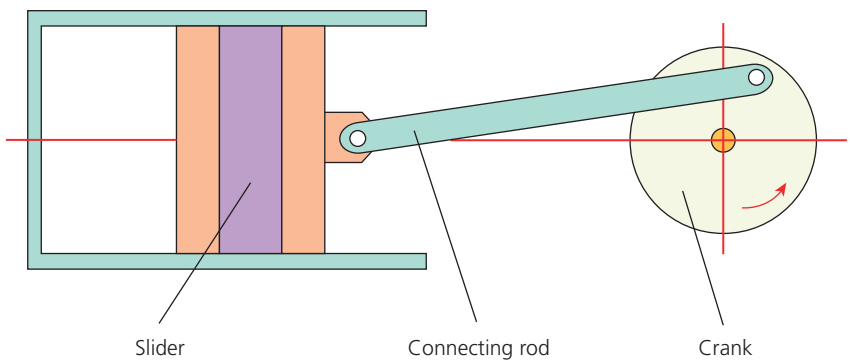


Figure 1.15 A crank and slider mechanism

1.5.8 Gear types

REVISED

- In a simple two **gear train**, a larger gear driving a smaller gear results in increased speed but decreased torque. A smaller gear driving a larger gear achieves the opposite.

Gear train: a collection of spur gears that transmit rotary motion and torque.

$$\text{rpm of the driven gear} = \frac{\text{rpm of the driver gear}}{\text{Gear ratio}}$$

- **Compound gear trains** are created when one or more of the shafts holds two gears. They are used when large increases in speed or torque are needed and when space is limited.
- An **idler gear** is placed between two gears to ensure that they have the same direction of motion.
- **Bevel gears** are used to change the direction of a shaft's rotation, usually through 90 degrees.
- **Rack and pinions** change rotary motion to linear motion. The teeth on the circular pinion gear mesh with those on the rack. As the pinion gear turns, the rack moves in a straight line.

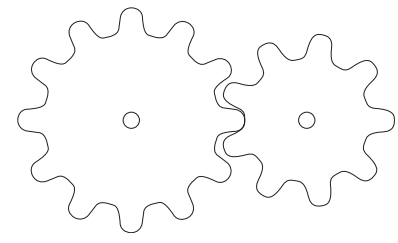


Figure 1.16 A simple gear train

Now test yourself

TESTED

1 State what is meant by the term 'rotary movement'. (1)
2 Give an example of a class 2 lever. (1)
3 Name **three** types of follower in a cam and follower mechanism. (3)
4 Describe the function of a crank and slider mechanism. (2)
5 Describe the purpose of an idler gear. (2)

1.6 Electronic systems

1.6.1 Sensors

REVISED

- Sensors are input devices. They take signals from the environment around them, such as light and temperature levels, and turn them into electronic signals, such as current, voltage or resistance.
- **LDRs** change the light level that is detected into a resistance. The resistance decreases as the brightness increases. LDRs are often used in outdoor street lamps and night lighting.
- **Thermistors** change the temperature level that is detected into a resistance. Thermistors are often used where it is important to know the temperature, such as inside a greenhouse or a refrigerator.

Thermistor: a sensor input that can detect changes in temperature levels.



Figure 1.17 A light-dependent resistor (LDR)



Figure 1.18 A push to make switch

1.6.2 Control devices and components

REVISED

- **Switches** are used to either 'make' (allow current to flow through) or 'break' (not allow current to flow through) a circuit. An example is a light switch that turns a lamp 'on'.
- Switches are described by either the number of poles and throws that they have, or by their function. For example, a push to make switch 'makes' a circuit when it is pressed.
- Other examples of switches include rocker switches, toggle switches, tilt switches, reed (magnetic) switches and micro switches.
- **Transistors** are used as either an electronic switch or a current amplifier. A transistor has three leads called the base, collector and emitter. When the voltage at the base reaches approximately 0.6 V, current is allowed to flow between the collector and the emitter.
- **Resistors** are used to reduce the flow of current in a circuit. The higher the value of a resistor, the more resistance it has. The amount of resistance is measured in ohms, or Ω.

Switch: a component that can be used to either 'make' or 'break' a circuit.

Transistor: a component that acts as either an electronic switch or a current amplifier.

Resistor: a component that reduces the flow of current.

1.6.3 Outputs

REVISED

- Outputs take electronic signals and turn them back into 'real world' signals, such as light or sound.
- **Buzzers** use an internal oscillator to produce sounds at different frequencies when current flows through them. They are often used in alarms, warning systems and electronic children's toys.

Buzzer: an output device that produces sound when current flows through it.

1.17 Developing, communicating, recording and justifying design ideas

1.17.1 Using a range of communication techniques and media

REVISED

Sketching

- **Freehand sketching** is a method of presenting design ideas on paper without the use of drawing aids or formal drawing conventions. It is often used to produce early ideas for products or as a tool for ideas generation. Sketching can be done in either 2D or 3D.
- Annotations are used to give further details and information about the ideas that have been sketched. This could cover areas such as:
  - the inspiration behind each design
  - the good and bad points
  - manufacturing methods that could be used to make each idea
  - how well each design meets the requirements of the **end user** and/or design specification.

Freehand sketching: method of presenting design ideas on paper without the use of drawing aids or formal drawing conventions.

End user: the person who will purchase or use the finished product.

Cut and paste techniques

Cut and paste techniques are used to make collages. Pictures and images are cut from different sources, such as newspapers, magazines and photographs, and glued on to paper to create an art form.

Digital photography/media

Digital technologies are increasingly being used in the design of products. For example, taking photos of iterative models to record each stage of development, or videos of testing to demonstrate how a design works. These can be shared online with the end user or other potential customers.

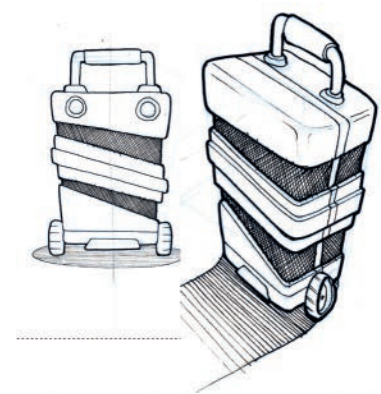


Figure 1.33 3D sketching

3D models

The use of **3D models** allows design ideas to visualised, handled and tested. Models allow the designer to spot any errors or areas for improvement before money is spent on more expensive materials. They can also be shared with the end user to gain their feedback before the final product is manufactured.

3D model: 3D version of a design idea made using cheap materials, for example card and Styrofoam™, used to check how it would look and function.

Isometric projection: method of drawing objects in 3D by drawing the sides at a 30-degree angle.

Isometric and oblique projections

- These are methods of drawing objects in 3D.
- In **isometric projections** the sides are drawn at a 30-degree angle.
- Oblique projections are simpler to produce than isometric, but not as realistic. The front side of the object is drawn first and lines are then drawn backwards at 45 degrees. The object is inclined either to the left or to the right.